

ICT2217 Network Security
Lab Exercise 1.3: Double-Tagging Attack and Defense

Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Conduct and defend against double-tagging attack

Required Hardware

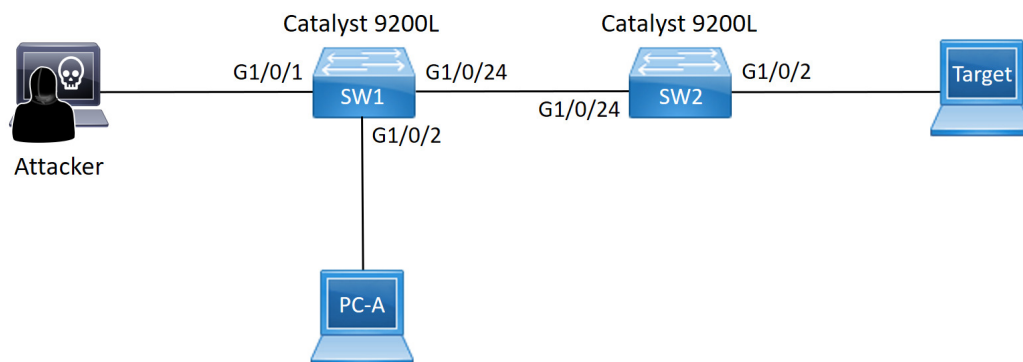
- 1 x Rack of Cisco networking devices
- 1 x Box of cables containing
 - USB-to-DB9 serial adapter + DB9-to-RJ45 console cable
 - USB-to-RJ45 console adapter
 - Ethernet cables
- 3 x Laptops

Required Software

- Tera Term <https://teratermproject.github.io/index-en.html>
- Kali Linux Live Boot on USB drive

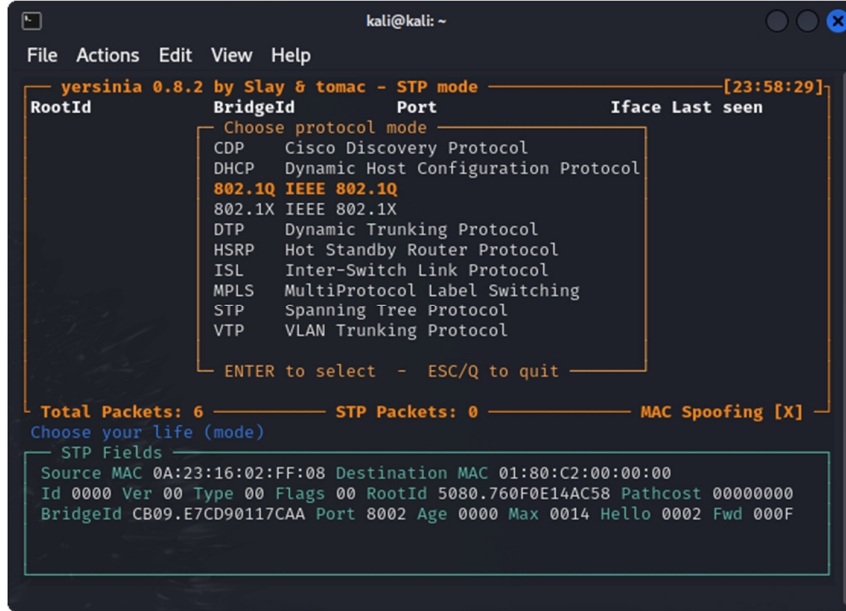
PART 1: Conducting Double-Tagging Attack using Yersinia

1.1 Set up the network topology as shown:



- 1.2 Configure interface G1/0/24 connecting switches SW1 and SW2 as trunk with default native VLAN 1, and SW2 G1/0/2 connecting target PC as VLAN 2. Leave SW1 G1/0/1 and G1/0/2 connecting Attacker's PC and PC-A in default VLAN 1.
- 1.3 Boot up attacker's PC in Kali Linux. The target PC and PC-A may be in Windows or Kali Linux.
- 1.4 Assign suitable IP addresses to the target PC and PC-A manually.

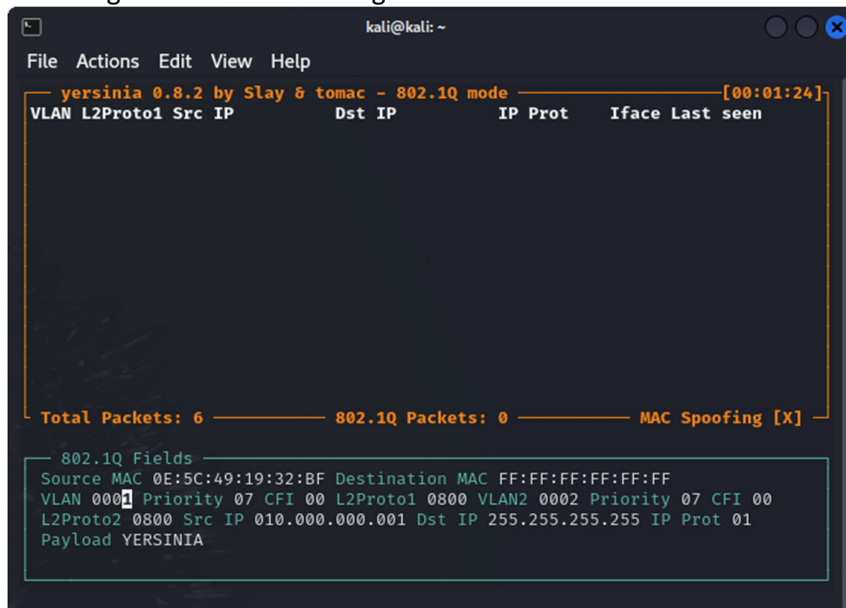
- 1.5 Start Wireshark at the attacker's PC to get ready to capture the double-tagging frame you are going to send out.
- 1.6 In addition, start Wireshark at the target PC to get ready to capture the double-tagging frame to be sent by the attacker.
- 1.7 When ready, start Yersinia in interactive mode at the attacker's PC.
- 1.8 Press key 'g' to go to protocol screen and select IEEE 802.1Q.



```

kali@kali: ~
File Actions Edit View Help
-- yersinia 0.8.2 by Slay & tomac - STP mode [23:58:29]
RootId BridgeId Port Iface Last seen
Choose protocol mode
CDP Cisco Discovery Protocol
DHCP Dynamic Host Configuration Protocol
802.1Q IEEE 802.1Q
802.1X IEEE 802.1X
DTP Dynamic Trunking Protocol
HSRP Hot Standby Router Protocol
ISL Inter-Switch Link Protocol
MPLS MultiProtocol Label Switching
STP Spanning Tree Protocol
VTP VLAN Trunking Protocol
ENTER to select - ESC/Q to quit
Total Packets: 6 STP Packets: 0 MAC Spoofing [X]
Choose your life (mode)
STP Fields
Source MAC 0A:23:16:02:FF:08 Destination MAC 01:80:C2:00:00:00
Id 0000 Ver 00 Type 00 Flags 00 RootId 5080.760F0E14AC58 Pathcost 00000000
BridgeId CB09.E7CD90117CAA Port 8002 Age 0000 Max 0014 Hello 0002 Fwd 000F
  
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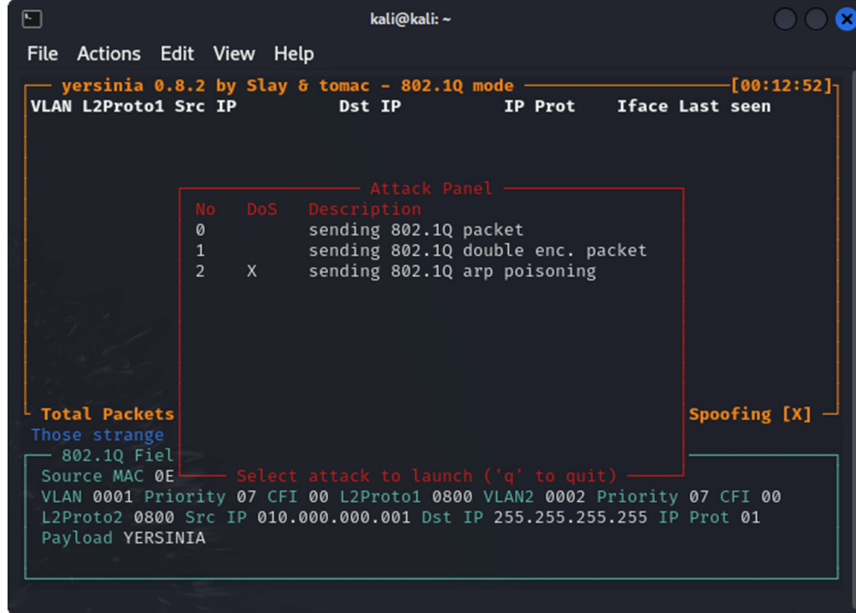
- 1.9 Press key 'e' and use arrow keys to navigate to the VLAN fields to edit the values if needed, then press ENTER to end the edit mode. For this lab, we are using the default outer tag VLAN 1 and inner tag VLAN 2.



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kali@kali: ~
File Actions Edit View Help
-- yersinia 0.8.2 by Slay & tomac - 802.1Q mode [00:01:24]
VLAN L2Proto1 Src IP Dst IP IP Prot Iface Last seen
Total Packets: 6 802.1Q Packets: 0 MAC Spoofing [X]
802.1Q Fields
Source MAC 0E:5C:49:19:32:BF Destination MAC FF:FF:FF:FF:FF:FF
VLAN 0001 Priority 07 CFI 00 L2Proto1 0800 VLAN2 0002 Priority 07 CFI 00
L2Proto2 0800 Src IP 010.000.000.001 Dst IP 255.255.255.255 IP Prot 01
Payload YERSINIA
  
```

- 1.11 Press key 'x' to eXecute attack panel.



```

kali@kali: ~
File Actions Edit View Help
yersinia 0.8.2 by Slay & tomac - 802.1Q mode [00:12:52]
VLAN L2Proto1 Src IP Dst IP IP Prot Iface Last seen

Attack Panel
No DoS Description
0 sending 802.1Q packet
1 sending 802.1Q double enc. packet
2 X sending 802.1Q arp poisoning

Total Packets
Those strange
802.1Q Fiel
Source MAC 0E
Select attack to launch ('q' to quit)
VLAN 0001 Priority 07 CFI 00 L2Proto1 0800 VLAN2 0002 Priority 07 CFI 00
L2Proto2 0800 Src IP 010.000.000.001 Dst IP 255.255.255.255 IP Prot 01
Payload YERSINIA
  
```

- 1.12 Finally, press key '1' sending 802.1Q double enc. packet to launch the attack.

WARNING: Disable Wi-Fi to prevent attacking other networks through Wi-Fi which may get you into trouble.

- 1.13 Stop the Wireshark at attacker's PC. Can you find the double-tagging frame that was sent out?
- 1.14 Stop the Wirehsark at the target PC. Can you find the frame sent by the attacker? Why or why not? Explain.
- 1.15 If you could not capture the frame at the target PC sent by the attacker in Step 1.12, troubleshoot the problem so that you are able to capture it successfully.

PART 2: Defending against Double-Tagging Attack

- 2.1 Implement defense against double-tagging attack as discussed.
- 2.2 Which VLAN do you expect the the double-tagging frame will go to after your defense?
- 2.3 Restart Wireshark at attacker's PC to get ready to capture the double-tagging frame you are going to send out.
- 2.4 Restart Wireshark at target PC to get ready to capture any frame to be sent by the attacker.

- 2.5 In addition, start Wireshark at PC-A to verify whether it will receive the frame sent by the attacker.
- 2.6 Launch Yersinia to conduct double-tagging attack again.
- 2.7 Stop Wireshark at attacker's PC. Can you find the double-tagging frame being sent out?
- 2.8 Stop Wireshark at target PC. Can you find the frame sent by the attacker now?
- 2.9 Stop Wireshark at PC-A. Can you find the frame sent by the attacker?
- 2.10 Do you think the recommended defenses are effective against double- tagging attack? Explain.